

Topology Optimization
Hub Carrier optimization

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A.Y. 2024/25



POLITECNICO
MILANO 1863

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1 Introduction

The design of lightweight structures is a crucial aspect in the automotive industry. In the following, a topology optimization of the hub carrier for an ultra-efficient lightweight battery-electric vehicle is presented. In particular, the vehicle considered for this work is *Asteria* (see Figure 1), developed by the *Green MECC* team at Politecnico di Milano for the *Shell Eco-marathon* competition.



Figure 1: Asteria vehicle [2].

During the vehicle motion, the main sources of energy dissipation are mass (inertia), tire rolling resistance, and aerodynamic drag. Moreover, the vehicle mass has the most significant impact on energy consumption, emphasizing the importance of lightweight design.

Project goals The goal of this project is to optimize the hub carrier by maximizing structural stiffness while imposing a mass constraint. As will be discussed along the document, different load scenario and constraints are considered to ensure the hub carrier is able to withstand the forces generated during the competition. Some constraints will also be imposed to ensure the manufacturability of the final design.

Tools The optimization process will be carried out using some of the software package provided by the **Altair HyperWorks** suite [1]. In particular, we opted for the following tools:

- **HyperMesh**: preprocessing and meshing phase;
- **OptiStruct**: topology optimization;
- **HyperView**: post-processing and visualization.

Finally, the smoothing of the optimized geometry is performed using **Altair Inspire**.

2 Optimization problem

As already mentioned in the introduction, the goal of this project is to optimize the hub carrier by maximizing structural stiffness while minimizing mass.

The hub carrier (shown as one of the components in Figure 2) has three main functions:

- It closes the kinematic loop made by the two control arms;
- It receives the inputs from the steering rod and transfers them to the wheel;
- It bears the static and dynamic loads coming from the road and the brake caliper.

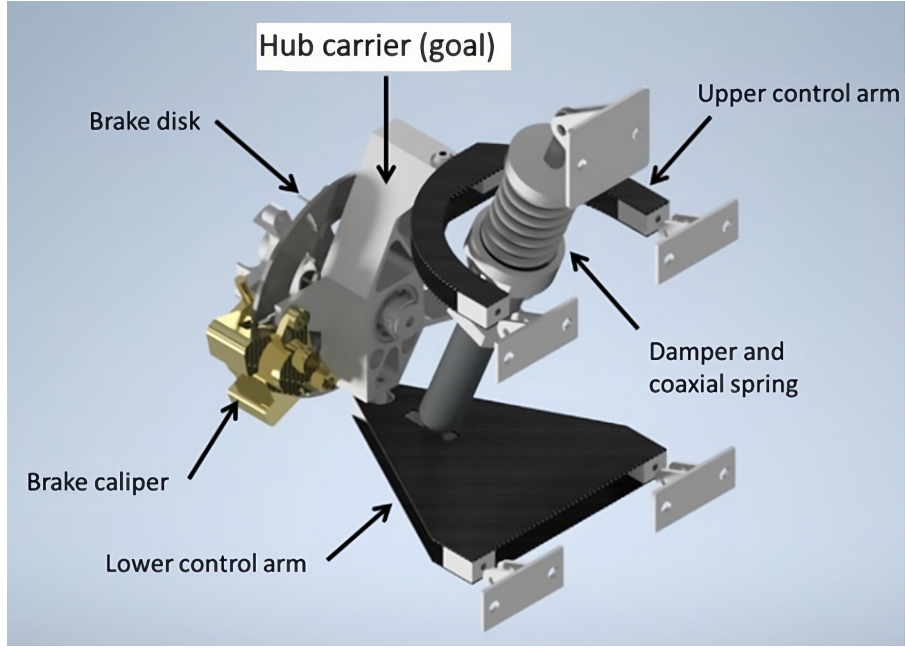


Figure 2: Schematic representation of the front wheel assembly. The hub carrier is indicated by a proper label.

2.1 Design domain

As a first approximation, the design domain is defined as the volume enclosed by the hub carrier's outer surface. This choice is made to avoid the definition of complex initial geometries and not restrict the optimization algorithm to a specific pre-designed shape.

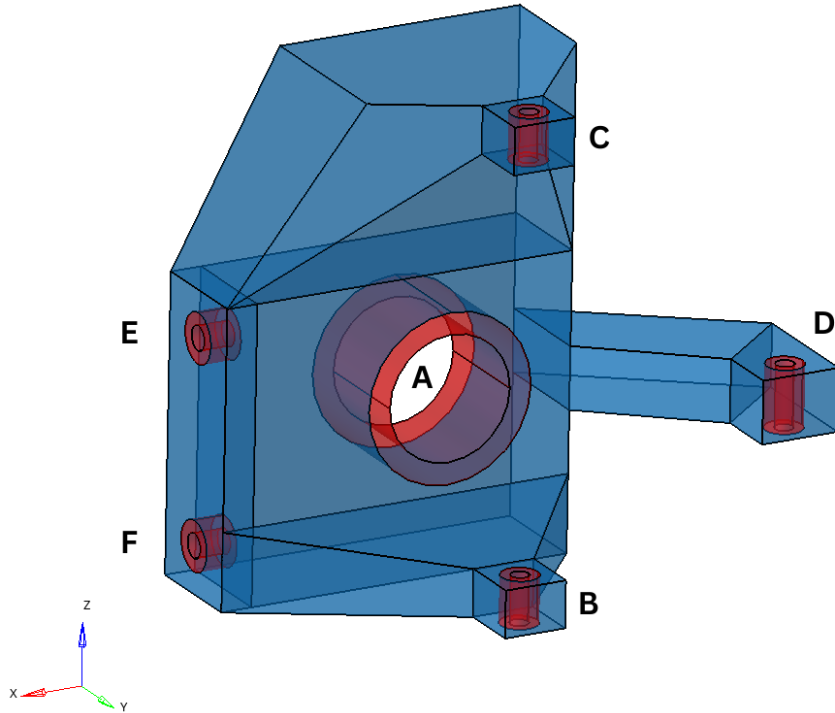


Figure 3: Initial domain definition.

In Figure 3, the design domain is shown in blue, while the non-design domains (i.e., the hub carrier's connecting points with the rest of the vehicle) are shown in red. Connection points are highlighted via the labels $A \dots F$. The function of each connection point is as follows:

- Point A : connection with the wheel spindle, all the loads coming from the wheel are transferred to this point;

- Point *B*: connection with the lower control arm and the damper, all the vertical loads are transferred to this point;
- Point *C*: connection with the upper control arm;
- Point *D*: connection with the steering rod;
- Points *E* and *F*: connection with the brake caliper.

Based on the function of each connection point, the following boundary conditions are applied to the design domain:

- Point *B*: full 3D pin constraint (i.e., free rotation but no translation);
- Point *C*: constrained in the $\vec{x}$ and $\vec{y}$ directions;
- Point *D*: constrained in the $\vec{y}$ direction;

2.2 Load cases

In the following, a brief description of the load cases is provided.

Pure rolling The hub carrier is subject only to the vertical load coming from the vehicle's weight. The load is applied at point *A* and has a magnitude of $F_z = 400[N]$.

Braking Due to the braking action, the load is transferred from the rear to the front axle and a horizontal force is also generated. Loads are applied to point *A* with a magnitude of $F_z = 1300[N]$ and $F_x = -200[N]$, and points *E* and *F* with a magnitude of $F_z = -350[N]$ each.

Internal turn Due to inertia, the vehicle is subject to a lateral acceleration which generates a lateral force on the wheel and a moment on the hub carrier. Loads are applied to point *A* with a magnitude of $F_y = -200[N]$, $F_z = 300[N]$, $M_x = -60[Nm]$ and $M_z = 4[Nm]$.

External turn Loads are essentially the same as the internal turn, but with opposite signs, except for the vertical force in *A* which became $F_z = 500[N]$.

Internal bump/corners In case of a bump or a corner, the wheel is subject to string loads due to high accelerations involved in the phenomena. We can consider loads applied to point *A* with a magnitude of $F_x = 500[N]$, $F_y = 500[N]$, $F_z = 1000[N]$ and $M_x = 150[Nm]$.

External bump/corners Loads are essentially the same as the internal bump/corners, but with opposite signs, except for the vertical force in *A* that remains $F_z = 1000[N]$.

2.3 Material properties

In order to minimize the mass of the hub carrier, the material choice has to be carefully made. In particular, the use of steel is not recommended due to its high density which would lead to a heavy component. For this reason, the material chosen is *Aluminum 6061 T6*, which has a density of $\rho = 2700[kg/m^3]$ and a Young's modulus of $E = 70[GPa]$. The yield strength is $\sigma_{YS} = 275[MPa]$ and the Poisson's ratio is $\nu = 0.33$.

2.4 Objectives and constraints

The formal definition of the topology optimization problem can be given as the minimization of the weighted compliance C for the six load cases described in Section 2.2 subjected to the following constraints:

- Camber stiffness: $K_c \geq 160 \cdot 10^3[Nm/rad]$;
- Toe stiffness: $K_t \geq 84 \cdot 10^3[Nm/rad]$;
- Mass fraction: $m_{opt} \leq 0.225 \cdot m_0$;
- Manufacturing constraints: constant cross-sections along the $\vec{y}$ direction (i.e., extrusion like shapes).

The optimization variables are the density of each element in the design domain.

3 Results

The final topology optimization results are shown in Figures 4 and 5.

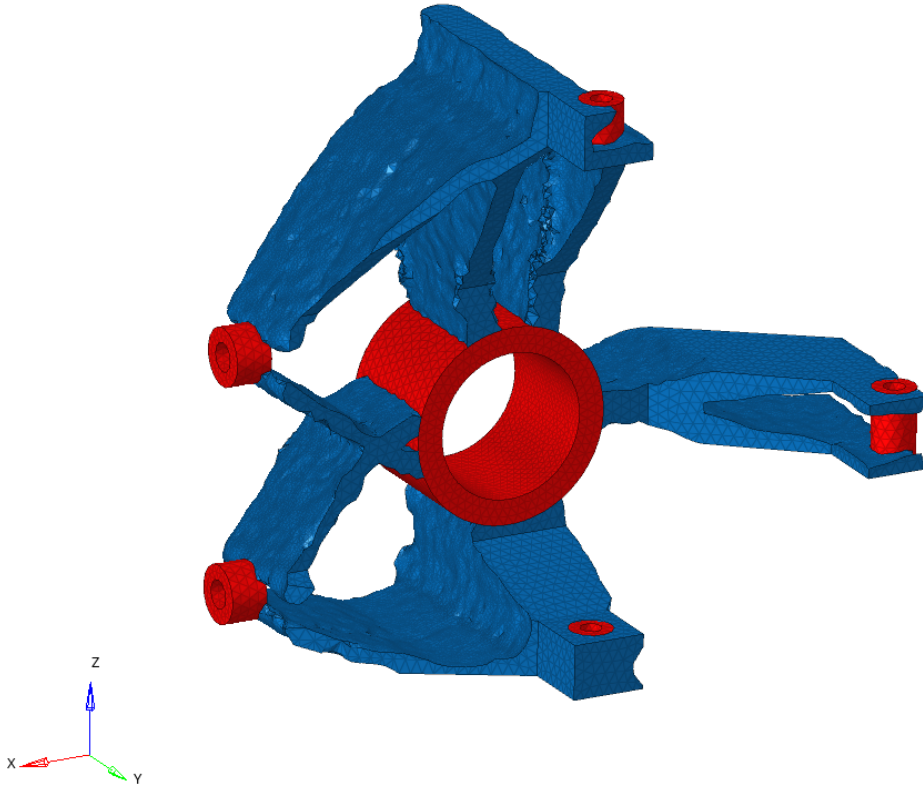


Figure 4: Optimized design.

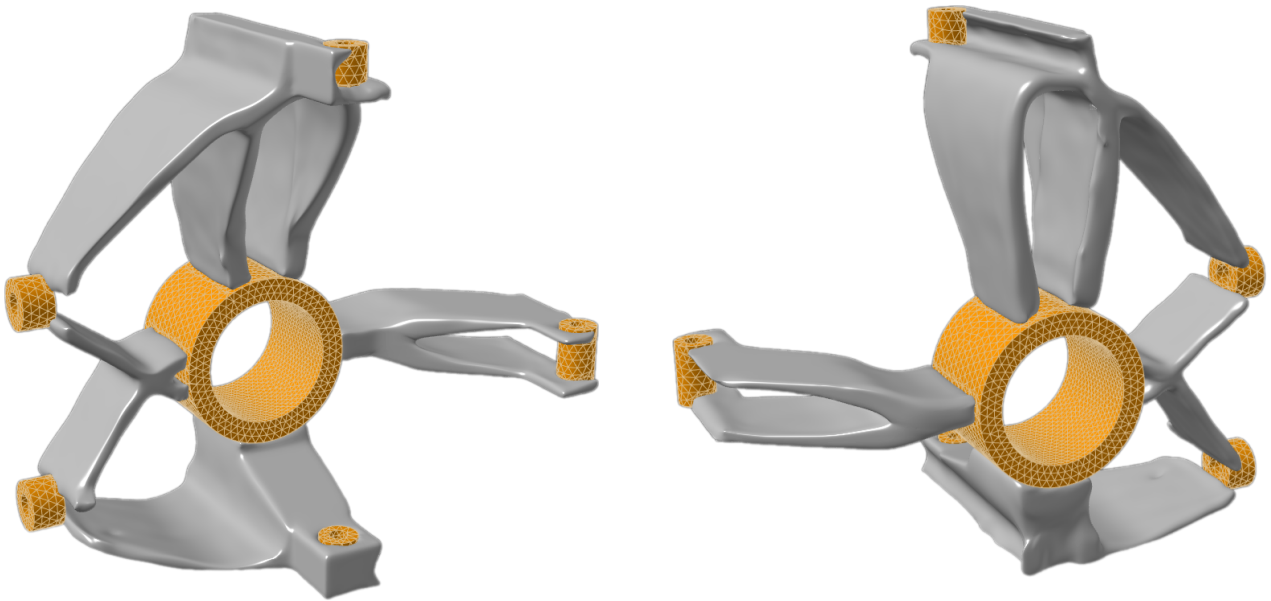


Figure 5: Smoothed design.

Due to limitation in the number of elements imposed by the student version of the software, the meshing element has been chosen to be tetrahedral with a size of $3.0[mm]$ on both the design and non-design domain. The optimization process converged after 47 iterations with a final weighted compliance of $224.15[J]$ and a

final mass of $0.45[kg]$ (initial mass was $1.98[kg]$). Both the constraints imposed on the stiffness are respected (stiffness computed as the inverse of the maximum rotation at point A in presence of a unitary moment):

- Camber stiffness: $K_c = 1/6.26 \cdot 10^{-6}[Nm/rad] = 159.7 \cdot 10^3[Nm/rad]$;
- Toe stiffness: $K_t = 1/1.14 \cdot 10^{-5}[Nm/rad] = 87.7 \cdot 10^3[Nm/rad]$.

Given the imposed high stiffness constraints, we can expect that also the dissipation of energy from the vehicle due to out-of-plane deformations will be minimized, thus improving the overall vehicle performance.

3.1 Finite Element Analysis

The final design has been analyzed using a Finite Element Analysis (FEA) to verify the stiffness and the stress distribution. The results are shown in the following figures.

From Von Mises stress distribution, it is possible to observe that the hotspots is located at the interface between the lower control arm and the main body of the hub carrier. Moreover, the most stressed condition appears to be the braking load cases, with a maximum stress of $37.4[MPa]$ which is below the yield strength of the material.

For deformations, a scale factor of 500 has been applied for visualization purposes.

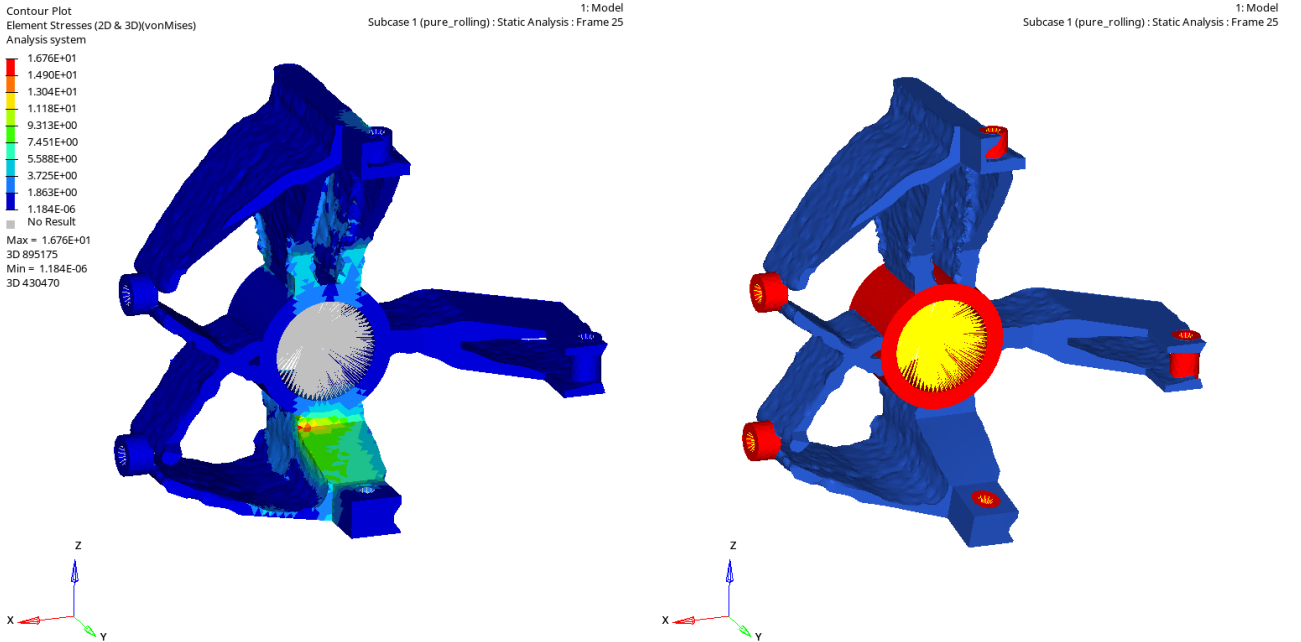


Figure 6: Pure rolling.

Contour Plot
Element Stresses (2D & 3D)(vonMises)
Analysis system

3.735E+01
3.320E+01
2.905E+01
2.490E+01
2.075E+01
1.660E+01
1.245E+01
8.304E+00
4.155E+00
6.263E-03
No Result

Max = 3.735E+01
3D 567880
Min = 6.263E-03
3D 430552

1: Model
Subcase 2 (braking) : Static Analysis : Frame 25

1: Model
Subcase 2 (braking) : Static Analysis : Frame 25

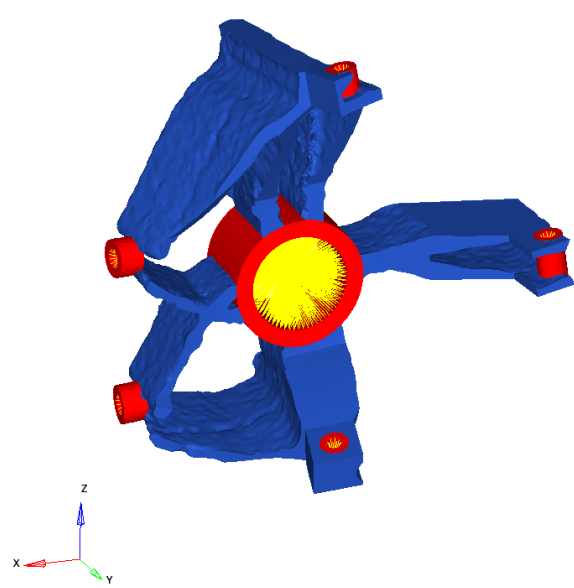
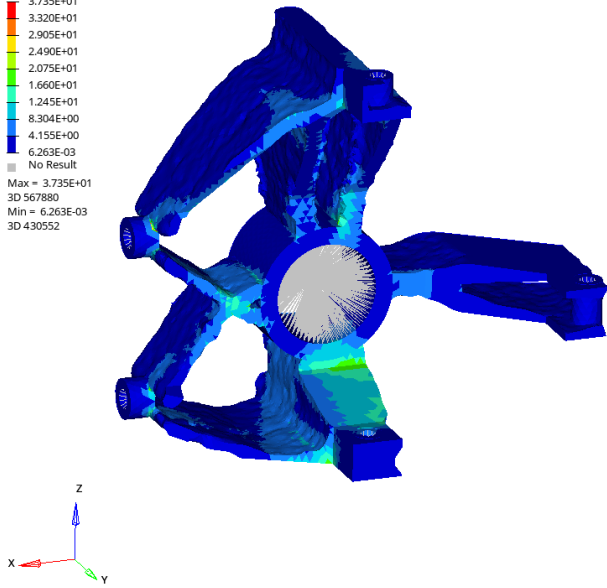


Figure 7: Braking.

Contour Plot
Element Stresses (2D & 3D)(vonMises)
Analysis system

1.183E+01
1.052E+01
9.203E+00
7.888E+00
6.574E+00
5.259E+00
3.944E+00
2.629E+00
1.315E+00
2.251E-06
No Result

Max = 1.183E+01
3D 895175
Min = 2.251E-06
3D 430552

1: Model
Subcase 3 (internal_turn) : Static Analysis : Frame 25

1: Model
Subcase 3 (internal_turn) : Static Analysis : Frame 25

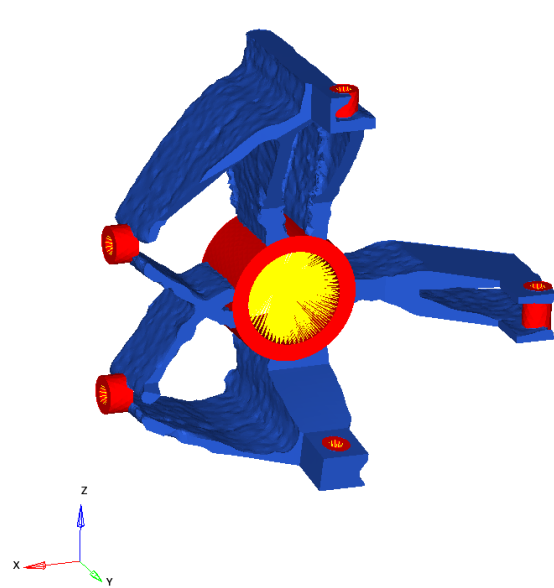
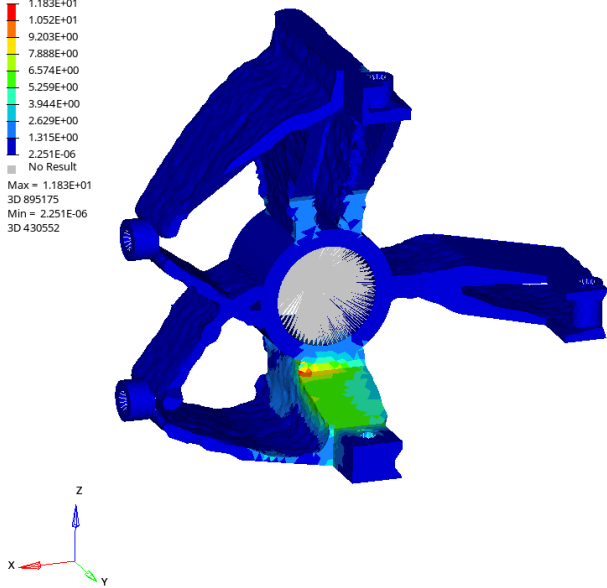
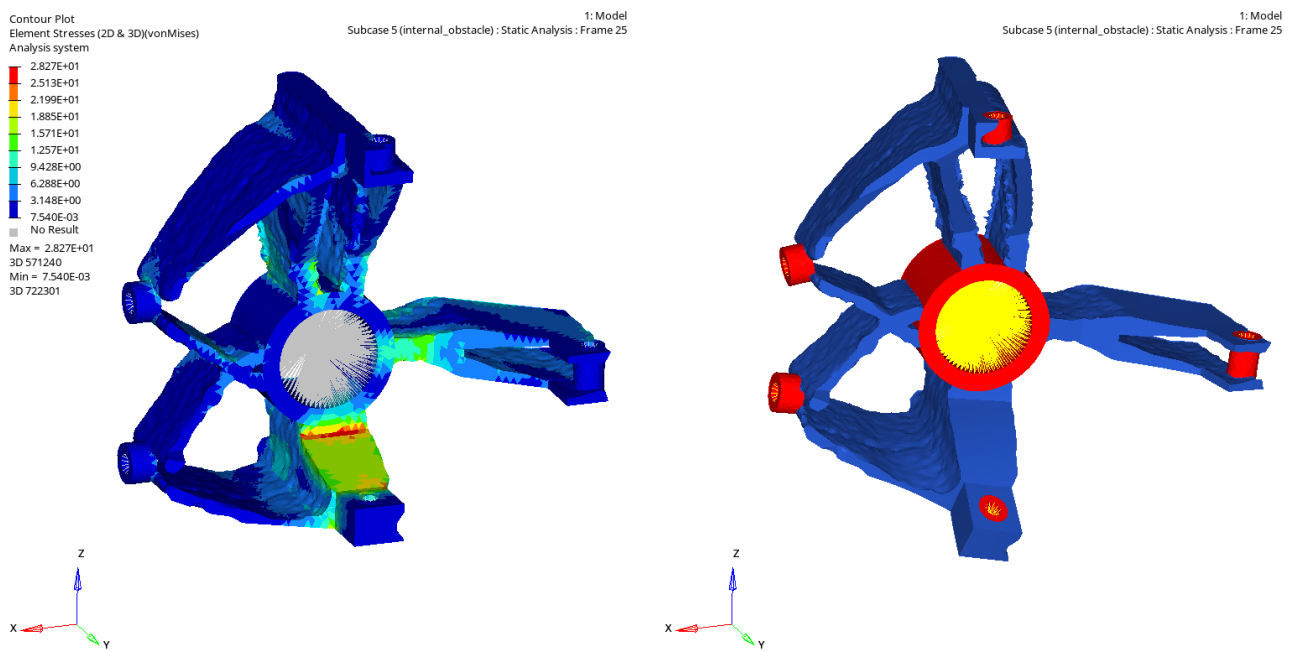
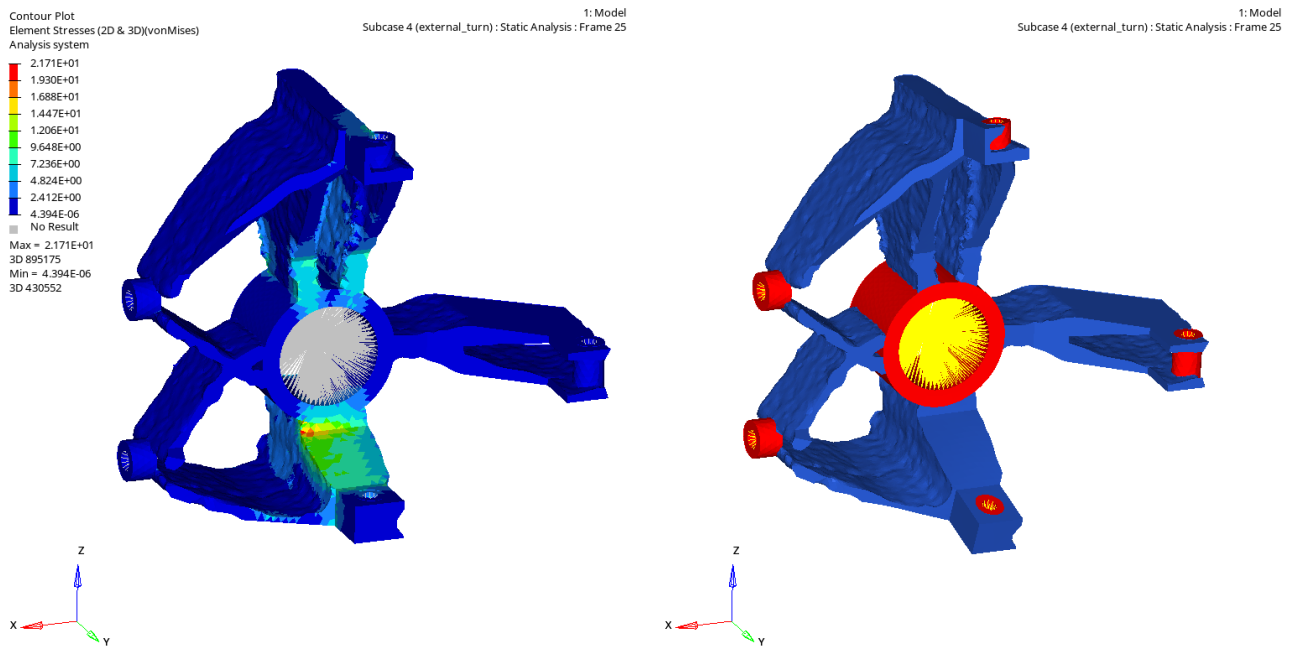
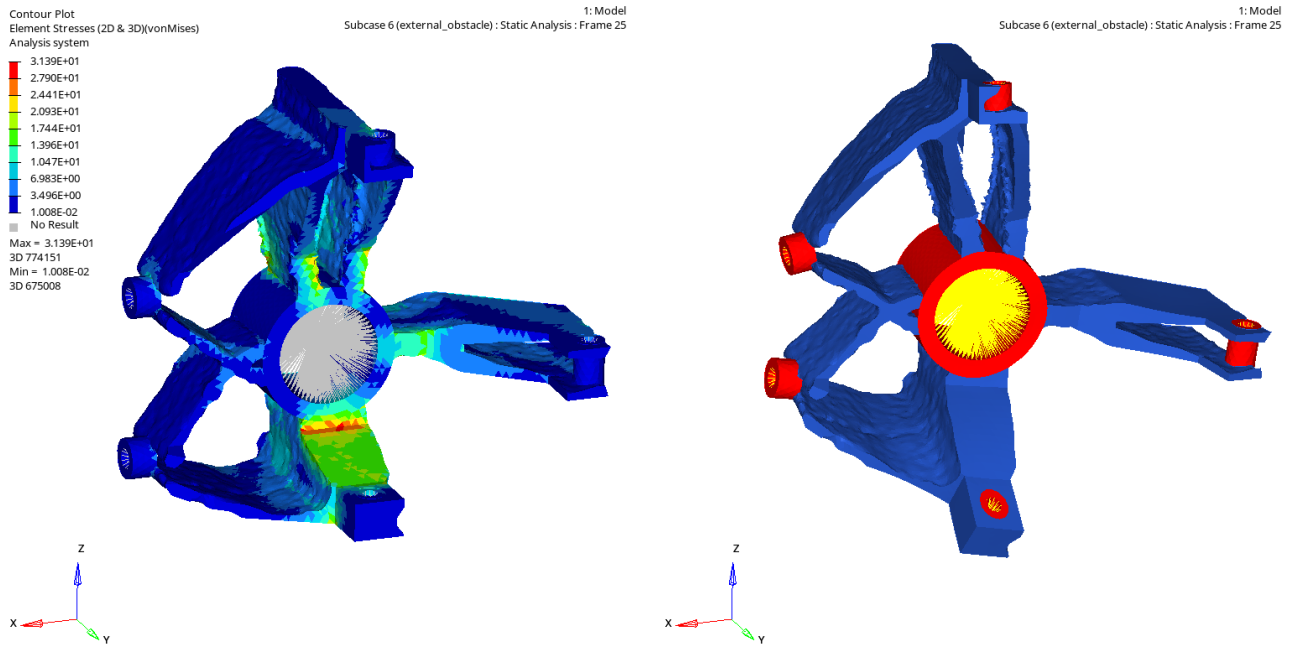


Figure 8: Internal turn.





References

- [1] Altair Engineering Inc. Design and simulation platform — altair hyperworks. <https://altair.com/altair-hyperworks>, 2025. [Online; accessed 26-January-2025].
- [2] Team Green Mecc. Team green mecc - shell eco-marathon - politecnico di milano. <https://sem.mecc.polimi.it/index.html>, 2025. [Online; accessed 26-January-2025].